FishAI v1.0: When Best-Response Adaptation Is Worth Less Than Nothing

Allen Hsieh
FishAI project

August 2026

Abstract

FishAI v0.5 measured a nine-style roster for the `us54` dialect of Literature (Canadian Fish) and found one style, Punter, dominant. FishAI v1.0 asks the natural follow-up: does an agent that identifies its opponents' styles online and best-responds beat committing to the best static style? The agent is built honestly — fourteen behavioural features certified by the public log alone, calibrated diagonal-Gaussian posteriors per opponent seat, best response over the measured 9×9 counter table, delegation to the unchanged style-conditioned engine, all under determinism and statelessness constraints that force co-located seats to identical reads. The answer is a theorem plus a measurement. The theorem: the Punter row of the measured table dominates every column, and expected payoff is linear in the opponent posterior, so the warm best response is Punter under *every* belief — the adaptive policy provably degenerates to the dominant static style. The measurement: the degenerate policy is strictly *worse* than static Punter, because its warmup anchor pays a tax. The adaptive team lands below the paired Punter benchmark in all nine gauntlet cells (mean -0.0108; sharpest cell -0.0108 ± 0.0031, $z = -3.53$) and loses a truly-paired 24-composition mixed screen at -0.0136 ± 0.0043 ($z = -3.2$, seed-clustered). An oracle ablation prices perfect classification at exactly zero: handed the true styles, the agent plays move-for-move identical games. The classifier has real but bounded skill — 22.4% end-of-game top-1 against an 11.1% chance floor, 50.4% on Ghost, 5.6–12.9% inside a quadrangle of styles that diverge from the control on under 3% of decisions — and under dominance none of it can matter. Adaptation over this roster is worth less than nothing, and every knob that repairs the tax does so by collapsing the policy into the static dominant style. That is a statement about this meta-game, not about adaptation as an idea, and we state which conditions — intransitive rosters, off-roster opponents, rule-set shifts — would give the machinery something to do.

1 Introduction

The FishAI v0.5 tournament [6] ended with a static answer to a static question: among nine parameterized play styles sharing one full-strength inference engine, the completion-chasing Punter dominates. The obvious rejoinder, standard in every community that argues about styles, is dynamic: *a fixed style is a sitting target; a player who reads the table and adapts should beat any static commitment.* FishAI v1.0 was built to test that rejoinder under the same measurement discipline, with the question fixed before any experiment ran: *does adapting to the opponents' identified styles beat committing to the best static style?*

The v1.0 agent is the natural best-response architecture, built without shortcuts: an observation layer that extracts per-seat behavioural features the public log *certifies* rather than estimates (Section 4.1); a classifier that turns those features into a posterior over the nine roster styles, calibrated offline and honest about early-game ignorance (Section 4.2); a best-response layer that scores every candidate style against the posterior using the measured 9×9 counter table and delegates the chosen style to the unchanged v0.5 engine (Section 4.3); and structural honesty constraints — public information only, bit-level determinism, and statelessness via phase quantisation so co-located teammates reach identical reads (Section 4.4).

The result is a negative one, and it arrives in an unusually clean two-part form. First, **a one-paragraph theorem** (Section 5): the Punter row of the measured counter table dominates every column, and the agent's expected payoff is linear in its opponent posterior, so the warm best response is Punter under *every* belief — uniform, oracle-exact, or anything the classifier can produce — and the adaptive policy provably degenerates to the dominant static style. Second, **a measured warmup tax** (Section 7): degenerating to Punter would merely tie the static benchmark, but the measured agent does worse, because its warmup anchor (Balanced, played until the public log matures) is a below-Punter style played for 52–58% of the team's decisions. The tax is visible in all nine gauntlet cells (mean -0.0108), significant in the sharpest ($z = -3.53$), and confirmed at $z = -3.2$ in a truly-paired mixed-population screen. Two of the four pre-registered predictions — written down, verbatim, before the run — expected the tax to be negligible; both were refuted, and the refutation *is* the headline.

The paper's contributions: the architecture as an honest reference implementation of online style identification in Literature (Section 4); the dominance-degeneracy theorem and its corollary that perfect classification is worth exactly zero, confirmed by an oracle ablation whose per-deal deltas are identically 0.0000 (Sections 5, 7.3); the pre-registered suite and its verdicts (Sections 6–7); and a mechanism-level account of the warmup tax, with the observation that every remedy collapses the adaptive policy into the static dominant style — not a bug in the remedies but the finding itself (Section 8).

2 Related work

Opponent modelling — fitting a model of another agent's policy or type and responding to it — is surveyed by Albrecht and Stone [3]. In trick-taking card games, Rebstock et al. show that policy-based inference substantially improves state estimation in Skat [4]; the project's internal research synthesis [9] catalogues this family for Literature (its opponent-model exploiter, A14) and specifies a per-seat detection panel (S35, "the Reader") whose feature set anticipates the observation layer built here. This paper implements a deliberately narrow slice of that program — type identification over a known, closed roster — and measures what the slice is worth. The meta-game framing follows Balduzzi et al. [1]: the v0.5 matrix is the evaluation data, its maximum-entropy Nash places all mass on Punter, and α-Rank [2] concentrates likewise, so a best-response dynamic over that matrix has a unique fixed point; this paper measures the cost of travelling to it. The self-play caveat of Hu et al. [5] — behaviour optimised within a closed population need not transfer outside it — is inherited in Section 9. To our knowledge, resting on the synthesis's negative search report [9, 6], no prior quantitative study of online style adaptation exists for Literature.

3 Background: the game, the dialect, and the v0.5 roster

Literature is a six-player, two-team card game of public asks and hidden hands. Under the `us54` dialect — specified rule-complete in the rules document [10] and summarized in the companion paper [6] — the deck is 54 cards in nine half-suits ("sets" or "books") of six; each seat is dealt nine; asks require holding a card of the asked set and not the named card; declares happen out of turn in a deterministic declare window; *any* declaration error awards the set to the opponents; the game ends the moment a team is awarded five sets; ties are impossible. The engine is pure, deterministic TypeScript: one seed reproduces a byte-identical game.

FishAI v0.5 [6] defined nine parameterized styles over one shared full-strength inference engine — Balanced (the tuned control), Blitz and Punter (aggressive), Banker and Turtle (conservative), Hoarder (optionality), and Scout, Ghost and Archivist (information) — and measured their full-precision round robin (36 cells × 4,300 duplicate deal pairs; per-cell SE at most 0.005) [13]. The verdict was *dominant* — *Punter* (mean 0.5829, maximin 0.5190, cyclic energy 0.0112, exploitability 0.0025), with two measured caveats that matter here: the declare-threshold axis the roster's labels advertise is inert across the roster's entire range [7], so the nine styles differ from the control on only 0.39–2.89% of decisions [11]; and the Hoarder's theorized benefit, the contained-book turn-pass, was implemented and measured to be worth nothing [8]. The first caveat returns in Section 7.4 as a quantitative bound on what any public-log classifier can see.

4 The adaptive architecture

The v1.0 agent is a policy shape, not a new engine. At every decision it (i) observes the three opponent seats from the public log, (ii) classifies each into a posterior over the nine styles, (iii) best-responds by expected score rate over the measured counter table, and (iv) delegates: the chosen style is handed to the ordinary v0.5 engine, which decides exactly as for any static style at full strength [15]. Everything above the delegation boundary is a pure function of the seat's public view and the configuration.

4.1 Observation: features the public log certifies

The observation layer is a single $O(\text{events})$ pass over the public log producing one feature vector per seat. Its design rule is that every feature is an *exact* public observation, never an estimate: a hit locates the named card publicly at the asker until it moves again; an ask certifies that the asker held a card of the asked set (the only licence to ask) and proves they lacked the named card; a declaration's reveal lists the true holder of all six cards, so foreign and own-hand-only declares are exact events; and hand counts are *replayed* from the log rather than read from the view, which is what makes truncated-log evaluation honest — a prefix's replayed counts are correct at the truncation point while the view's end-of-game counts are not [15].

One omission shapes everything downstream: **declining to declare emits no event**. The declare window advances silently, so declare-window patience is invisible to any public observer, and — as Section 8.1 measures — the public log grows several times slower than the decision count.

Table 1 lists the fourteen features. Several are direct public signatures of roster mechanisms: `deadAskShare` is the contained-book turn-pass of [8] seen from outside; `foreignDeclareShare` and `declareBackload` are the Hoarder's audit trail; `ownHandOnlyShare` is the Turtle's gate made visible.

4.2 Classification: calibrated diagonal-Gaussian fingerprints

The classifier is deliberately the simplest model that can be honest: one diagonal Gaussian per style per game-length bucket, calibrated offline from 150 `us54` mirror games per style and committed with full provenance [14]. For an observed vector x and a style with per-feature mean μ and deviation σ, the score is the plain z-distance $\sum_i -\frac{1}{2}z_i^2$, $z_i = (x_i - \mu_i)/\sigma_i$, softmaxed over the nine styles. Four calibration decisions were made by measurement, not taste. *The Gaussian normaliser is dropped*: with sample SDs over a few hundred calibration vectors, $\sum_i -\ln\sigma_i$ is a large constant

Table 1: The fourteen classifier features, all rates or shares — length-invariant by construction (a first calibration over raw counts read every short game as "not the style"). Denominators are clamped at one.

Feature	Definition (per seat, over the observed log)
`hitRate`	hits / asks
`askDiversity`	distinct books asked into / asks
`sameBookRepeatRate`	consecutive same-book asks / (asks − 1)
`certainAskShare`	asks for a card publicly located at the target / asks
`deadAskShare`	asks for a card publicly located elsewhere / asks (a guaranteed miss)
`leakyAskShare`	asks into books with own-team located+certified count ≥ 4 / asks
`completionAskShare`	asks into books with own-team located count $= 5$ / asks
`missFewestShare`	share of misses where the target held strictly the fewest cards
`missMostShare`	...strictly the most
`askShare`	asks / observed events
`declareShare`	declares / observed events
`foreignDeclareShare`	declares of books the claimer held no card of / declares
`ownHandOnlyShare`	declares where every card was the claimer's / declares
`declareBackload`	mean declare position (event index / final index)

bonus for whichever style calibrated tightest — in the first calibration it handed the Ghost 40 of 60 mirror-Balanced reads — and without it the argmax at a style's own mean is that style. *Per-style variance is shrunk halfway to the pooled variance*, $\sigma'^2 = (\sigma^2_{\text{style}} + \sigma^2_{\text{pooled}})/2$: raw per-style SDs make the widest-variance style an outlier magnet, fully pooled SDs discard real information, and the blend beat both ends in calibration probes. *Checkpoint buckets*: a mid-game observation must be read against fingerprints taken at the same horizon, so the calibration stores buckets at $\{60, 120, 200, 300\}$ observed events plus a full-game bucket. *Sample-size damping*: the posterior is blended toward uniform by $\min(1, \text{asks}/12)$ — at zero observed asks the answer is exactly "I don't know", and full confidence requires a dozen of the seat's *own* choices, not merely a long game. Determinism is absolute: the same view and fingerprints produce identical output, ties resolving to the earlier fingerprint in roster order.

4.3 Best response over the measured counter table

The payoffs are the committed counter table P: P_{ij} is the duplicate-averaged score rate of style i against style j from the v0.5 full-precision matrix (matrix v2; diagonal 0.5 by duplicate symmetry) [13], reproduced in Table 2. Provenance travels with the table — the source artifact's digest is exported beside the numbers, pinned by tests, and enforced by the experiment harness (Section 6.4).

For opponent-seat posteriors q_1, q_2, q_3 over the styles, the agent scores every candidate row i:

$$E(i) \;=\; \frac{1}{3}\sum_{k=1}^{3}\,\sum_{s} q_k(s)\,P_{is} \;=\; \sum_{s}\bar{q}(s)\,P_{is}, \qquad \bar{q} = \tfrac{1}{3}(q_1 + q_2 + q_3), \tag{1}$$

adds a hysteresis bias β (`switchMargin`, default 0.01) to the anchor style's entry — a candidate must *beat* the anchor, not merely tie it — and plays the argmax. During *warmup* (fewer than `warmupEvents` = 40 observed events) the anchor (default Balanced, the control) plays unconditionally: an early posterior is mostly the damping prior, and a best response to ignorance is noise.

The chosen style is then *played* by the unchanged v0.5 engine at full strength: the adaptive layer selects among existing policies and adds no policy content of its own. A lab-only *oracle mode* substitutes a point-mass posterior with the true opponent styles, keeping warmup and phase behaviour identical, so the ablation of Section 7.3 isolates exactly one thing: the posterior.

Table 2: The measured counter table P (row's score rate vs. column), matrix v2 at 4,300 pairs per cell; diagonal 0.5 by duplicate symmetry; paired SEs 0.0036–0.0050; full precision in the committed table [13]. The Punter row exceeds every other row in every column.

	Bal	Bli	Pun	Ban	Tur	Hoa	Sco	Gho	Arc
Balanced	0.5	0.4962	0.4810	0.5270	0.6397	0.5624	0.6378	0.5549	0.5660
Blitz	0.5038	0.5	0.4802	0.5214	0.6427	0.5684	0.6356	0.5623	0.5669
Punter	**0.5190**	**0.5198**	**0.5**	**0.5469**	**0.6581**	**0.5795**	**0.6603**	**0.5867**	**0.5926**
Banker	0.4730	0.4786	0.4531	0.5	0.6171	0.5287	0.6056	0.5307	0.5410
Turtle	0.3603	0.3573	0.3419	0.3829	0.5	0.4198	0.5159	0.3957	0.4460
Hoarder	0.4376	0.4316	0.4205	0.4713	0.5802	0.5	0.5817	0.4780	0.5074
Scout	0.3622	0.3644	0.3397	0.3944	0.4841	0.4183	0.5	0.4434	0.4438
Ghost	0.4451	0.4377	0.4133	0.4693	0.6043	0.5220	0.5566	0.5	0.4944
Archivist	0.4340	0.4331	0.4074	0.4590	0.5540	0.4926	0.5562	0.5056	0.5

4.4 Honesty constraints

Three constraints are structural, inherited from v0.5 and extended. *Public information only*: the agent consumes `SeatView`, the exact public payload a human in the seat would see; no other hand is reachable by construction [6, 15]. *Determinism*: the style choice is a pure function of (view, configuration) — no clock, no randomness, no state between calls — so every reported game is reproducible from its seed. *Statelessness, via phase quantisation*: a stateful bot would remember its last choice and demand a margin to move off it; this bot owns no state, because two adaptive seats given the same public information *must* reach the same read — the roster's one-shared-engine discipline [11] would otherwise die at the adaptive layer. Hysteresis is therefore expressed statelessly, twice over: the memoryless anchor bias β, and *phase quantisation* — the choice may only change when $\lfloor \text{events}/30 \rfloor$ changes. In the implementation's own words: the posterior "is evaluated on the log cut to the last multiple of [30] events, so every view inside one phase — and every seat looking at it — evaluates the identical prefix and reaches the identical choice. A seat therefore cannot flip styles mid-phase, which is the behavioural point of hysteresis, without a single byte of remembered state" [15]. The warmup gate reads the same truncation, so the whole selection is a function of the phase alone; with the defaults the first warm phase begins at 60 observed events — the first multiple of 30 at or above 40.

5 The dominance-degeneracy theorem

One look at Table 2 gives the game away: the Punter row is the column-wise maximum everywhere. What that implies for the adaptive policy is one paragraph of linear algebra.

Theorem 1 (Dominance degeneracy) *Let $P \in \mathbb{R}^{n\times n}$ be the counter table and suppose row r weakly dominates every row: $P_{rs} \geq P_{is}$ for all styles i and s. Then for every triple of opponent posteriors (q_1, q_2, q_3), r maximizes the expected payoff $E(\cdot)$ of Eq. (1); if the dominance is strict in every column, the argmax is unique. Moreover, with the anchor bias — $E(a)$ replaced by $E(a) + \beta$ for the anchor $a \neq r$ — r remains the unique argmax whenever $\min_s (P_{rs} - P_{as}) > \beta$.*

Proof. By Eq. (1), E is linear in the mean posterior $\bar{q}$, which lies in the probability simplex. For any i, $E(r) - E(i) = \sum_s \bar{q}(s)\,(P_{rs} - P_{is})$ is a convex combination of non-negative numbers, hence ≥ 0; if every difference is strictly positive it is > 0, since $\bar{q}$ places mass somewhere. For the anchor, $E(r) - \big(E(a) + \beta\big) = \sum_s \bar{q}(s)\,(P_{rs} - P_{as}) - \beta \geq \min_s(P_{rs} - P_{as}) - \beta > 0$. □

The measured table instantiates the strict case: Punter's smallest margin over the best rival row is 0.0112 (Blitz row, Hoarder column), and its smallest margin over the Balanced anchor row is 0.0171 — above the default $\beta = 0.01$, so not even the anchor bias can rescue another choice. Hence:

Corollary 1 (The warm policy is static) *Under the committed table and defaults, the warm adaptive choice is Punter at every decision, under every posterior the classifier can produce and every oracle assignment. The adaptive policy is exactly: Balanced for the warmup phases, Punter thereafter.*

Corollary 2 (Perfect classification is worth exactly zero) *Classifier and oracle modes share the warmup and phase rules and differ only in the posterior. By Corollary 1 they delegate identically at every decision, so on the same seed they produce byte-identical games: the oracle ablation's per-deal delta is identically zero — not approximately, exactly.*

One statistical qualification, stated rather than buried. Dominance here is a property of the *committed point-estimate table*, which is the object the engine consults. As a claim about the true matrix it is strongly but not perfectly supported: of the 72 row-vs.-Punter-row comparisons (8 rival rows $\times$ 9 columns), 70 exceed even the Bonferroni-corrected two-sided bound $|z| > 2.773$ under a conservative independent combination of the paired cell SEs; the weakest, the Blitz row in the Hoarder column (0.0112), sits at $z = 1.86$. If the true matrix inverted that margin, the argmax could depend on beliefs concentrated on the Hoarder — and the value at stake would be bounded by roughly 0.011 in score rate. The implementation anticipates this honestly: its tests re-derive the argmax from the table rather than hard-coding Punter, so a future table with an intransitive cycle flows through the same code and simply makes the adaptation non-trivial [15].

6 Experimental design

6.1 Pre-registration

The committed counter table already implied Theorem 1; the experiments check the implication against play, not discover it. Four predictions were written into the suite's source before the run and travel verbatim inside the artifact [14]:

- **P1** The counter table's Punter row weakly dominates every column, so a warm adaptive team should match Punter's matrix-v2 row against every pure style within CI. Any detectable shortfall is the price of the warmup anchor (Balanced until $\sim$60 observed events).

- **P2** The adaptive-vs-Punter mixed-population delta should be ≈ 0: both teams best-respond to every composition with the same style once warm, so adaptation buys nothing a fixed Punter team does not already have.

- **P3** Oracle classification should add $\approx$ nothing: the best response is Punter regardless of the read, so a perfect read and a classifier read select the same play.

- **P4** Classifier top-1 accuracy should be good for Turtle, Ghost and Hoarder, and heavily confused inside the balanced/blitz/punter/banker quadrangle (the separability read: those four diverge from Balanced on under 1.3% of decisions [11]).

The verdict rules were committed alongside. P1 and P3 are nine simultaneous cells: a cell *rejects* when $|z|$ exceeds the Bonferroni-corrected two-sided 5% bound of 2.773; any rejection refutes; cells between 1.96 and the bound are named but do not refute (with nine cells, ∼0.45 land there by chance). P2 is one truly-paired comparison at $|z| \leq 1.96$. P4 was operationalized as three conditions on the end-of-game reads: mean top-1 over {Turtle, Ghost, Hoarder} ≥ 0.50; mean top-1 over the quadrangle ≤ 0.35; and, for every quadrangle style, at least half its errors landing inside the quadrangle. A refuted prediction is emitted as `refuted`, not massaged.

6.2 The five experiments

All experiments use duplicate deal pairs (each seeded deal played in both orientations, start seat rotating $i \bmod 6$), the paired estimator, and the v0.5 health discipline [12, 6]. The full suite is 125,600 games [14]:

1. **Gauntlet** — the adaptive team against each of the nine pure styles, at 4,300 pairs per cell on *matrix v2's exact seed list* (prefix `style-v1`), both orientations, same start seats. That choice is the experiment: every deal behind Punter's committed row is replayed by the adaptive team, making the Punter row a per-deal benchmark (cross-run caveat below). Every gauntlet game also records which style each adaptive decision delegated to, split warmup/warm by the engine's own phase rule — the recorded delegation *is* what was played.

2. **Adaptive mirror** — 400 pairs of adaptive-vs-adaptive. The policy is deterministic and identical on both teams, so the two orientations of a pair are literally the same game: the paired score must be exactly 0.5 with SE exactly 0. A plumbing symmetry check, and it passed (0.5000 ± 0.0000 over 800 games).

3. **Mixed-population screen** — 24 opponent compositions × 400 pairs on prefix `mixed-v1`, each played twice: the adaptive team, and a pure Punter team, on the same seeds, start seats and orientations. The headline is the pooled per-deal delta (adaptive − Punter), *truly paired within the run*; the harness verifies — not assumes — that both arms played identical (pair, orientation, seed, start-seat) sets. Because all 24 compositions replay the identical 400-seed list, the pooled SE is clustered by seed — per-deal deltas are averaged within seed before the SE is taken over the 400 seed-level means — so a deal's replays are never counted as independent evidence (per-composition SEs remain within-composition). The compositions are a fully specified deterministic sample: every 7th of the 165 lexicographic 3-multisets.

4. **Oracle ablation** — the nine gauntlet cells re-run at 400 pairs with the true opponent styles handed to the adaptive seats. The classifier arm is not re-run: the gauntlet's first 400 pairs *are* the classifier arm on identical seeds.

5. **Classifier accuracy** — all 36 unordered style pairings, 50 single games each on prefix `clsacc-v1`. Each finished game's public log is truncated at $\{40, 80, 150, 250\}$ events (a checkpoint scores only games whose log outlived it) plus the full log; top-1 is scored per seat against the seat's true style.

6.3 Paired benchmarks and the cross-run caveat

The gauntlet's benchmark is Punter's row of the committed matrix-v2 artifact [13], and the run's provenance block states the caveat verbatim [14]: "Per-deal: the gauntlet replayed the benchmark seed list exactly (both orientations, same start seats). The pairing is cross-run — per-game records are not joined — so the delta SE is the conservative independent combination, an upper bound on the true paired SE." The conservatism cuts one way: it makes the P1 test *harder* to refute, so the

Table 3: The gauntlet: adaptive team vs. each pure style, 4,300 pairs per cell on matrix v2's exact seed list, against Punter's committed row as a per-deal benchmark [14]. Δ = adaptive − benchmark, with the conservative cross-run SE; the pre-registered rejection bound is $|z| > 2.773$.

Opponent	Adaptive	SE	Punter row	SE	Δ	z
Balanced	0.5114	0.0027	0.5190	0.0036	-0.0076 ± 0.0045	−1.68
Blitz	0.5086	0.0040	0.5198	0.0042	-0.0112 ± 0.0058	−1.93
Punter	0.4892	0.0031	0.5000	0	-0.0108 ± 0.0031	**−3.53**
Banker	0.5393	0.0048	0.5469	0.0048	-0.0076 ± 0.0067	−1.12
Turtle	0.6485	0.0045	0.6581	0.0046	-0.0097 ± 0.0065	−1.49
Hoarder	0.5709	0.0039	0.5795	0.0042	-0.0086 ± 0.0057	−1.51
Scout	0.6512	0.0046	0.6603	0.0046	-0.0092 ± 0.0065	−1.40
Ghost	0.5691	0.0049	0.5867	0.0049	-0.0177 ± 0.0069	−2.55
Archivist	0.5778	0.0046	0.5926	0.0047	-0.0148 ± 0.0066	−2.24

refutation reported below is understated, not manufactured. The mixed screen and oracle ablation are paired within the run and carry no such caveat. One gauntlet cell is sharper than the rest: against Punter itself the benchmark is exactly 0.5 with SE 0 by duplicate symmetry.

6.4 Health gates and provenance

A run is void unless its counters are clean; for the reported run [14]: `illegalActions` 0, `cappedGames` 0, `invariantViolations` 0, `ties` 0, `voids` 0, `nonClinch` 0, and `distinctSeeds` equal to the expected pair count in every cell; the suite adds two pairing gates (the mixed arms' seed-set identity, and the oracle cells' identity with the gauntlet head), both passed. The artifact is committed at `src/lab/data/adaptive-results.json` with records digest `ce3316641a8f38fe`, engine commit `45ec9f3` (recorded `45ec9f3-dirty`), and the rules-document hash (`b4c7fef7...`) shared with v0.5; the 125,600 games took about 9.7 minutes of wall clock. Because an adaptive result is only as honest as the data it consulted, the artifact also carries the provenance of both committed calibrations — the counter table (from matrix v2, digest `0c3cb524a3534d4a`) and the fingerprints (`gen-fingerprints`, prefix `fingerprint-v1`) — and the assembly step *throws* if the benchmark's digest differs from the counter table's source digest: the row the gauntlet is read against and the table the agent best-responded with must come from the same matrix.

7 Results

7.1 P1: the gauntlet — refuted, by the warmup tax

Table 3 is the headline measurement. The adaptive team lands below the paired Punter benchmark in *all nine* cells — mean shortfall −0.0108 — and the sharpest cell rejects decisively: against Punter itself, where the benchmark is exactly 0.5 by symmetry, the adaptive team scores 0.4892 ± 0.0031, a delta of -0.0108 ± 0.0031, $z = -3.53$, past the pre-registered Bonferroni bound of 2.773. Two further cells sit between 1.96 and the bound (Ghost $z = -2.55$, Archivist $z = -2.24$). Under the committed rule, one rejection refutes: **P1 is refuted** — and since the cross-run SEs are conservative (Section 6.3; smallest detectable shortfall $\sim$0.0192 at the widest cell), the true significance is if anything understated. The prediction's own escape clause named the mechanism in advance — "any detectable shortfall is the price of the warmup anchor" — and Section 7.5 confirms the anchor is the only mechanism there is: once warm, the team delegated every single decision to Punter.

Table 4: The mixed screen: each composition (the opposing team's three styles) faced by the adaptive and pure-Punter teams on identical deals, 400 pairs per arm [14]. Δ = adaptive − Punter, per-deal paired.

Composition	Adpt.	Punt.	Δ	Composition	Adpt.	Punt.	Δ
bal+bal+bal	0.514	0.529	−0.0150 ±0.0101	pun+pun+hoa	0.533	0.545	−0.0125 ±0.0098
bal+bal+gho	0.576	0.573	+0.0038 ±0.0111	pun+ban+sco	0.543	0.561	−0.0188 ±0.0111
bal+bli+sco	0.561	0.584	−0.0225 ±0.0103	pun+tur+arc	0.588	0.608	−0.0200 ±0.0100
bal+pun+sco	0.540	0.565	−0.0250 ±0.0102	pun+sco+arc	0.551	0.561	−0.0100 ±0.0100
bal+ban+gho	0.561	0.574	−0.0125 ±0.0106	ban+ban+sco	0.583	0.573	+0.0100 ±0.0110
bal+hoa+hoa	0.551	0.563	−0.0113 ±0.0099	ban+tur+arc	0.599	0.611	−0.0125 ±0.0106
bal+gho+gho	0.544	0.566	−0.0225 ±0.0122	ban+sco+arc	0.584	0.589	−0.0050 ±0.0112
bli+bli+hoa	0.531	0.545	−0.0138 ±0.0102	tur+tur+gho	0.618	0.640	−0.0225 ±0.0107
bli+pun+hoa	0.523	0.540	−0.0175 ±0.0101	tur+sco+gho	0.604	0.616	−0.0125 ±0.0106
bli+ban+sco	0.538	0.569	−0.0313 ±0.0104	hoa+hoa+gho	0.581	0.604	−0.0225 ±0.0103
bli+tur+arc	0.584	0.604	−0.0200 ±0.0101	hoa+arc+arc	0.575	0.570	+0.0050 ±0.0097
bli+sco+arc	0.568	0.571	−0.0038 ±0.0102	gho+gho+gho	0.555	0.569	−0.0138 ±0.0126
Pooled per-deal delta (seed-clustered SE)				-0.0136 ± 0.0043 $(z = -3.18)$			

7.2 P2: the mixed-population screen — refuted, at $z = -3.2$

The mixed screen removes both the pure-team simplification and the cross-run caveat: 24 opponent compositions, each faced by the adaptive team and by a pure Punter team on identical deals within the same run. The pooled per-deal delta is -0.0136 ± 0.0043 ($z = -3.18$; 95% CI $[-0.0220, -0.0052]$) over 24×400 paired deals, with the SE clustered by seed (Section 6.2): the 24 compositions replay one 400-seed list, so the effective sample is 400 seed-level deltas, not 9,600 deltas counting each deal's replays as independent. **P2 is refuted**, at three standard errors. Per composition (Table 4), 21 of 24 deltas are negative; the three positive ones (+0.0038 to +0.0100) are each within one standard error of zero. Against heterogeneous opposition — where a read could at least in principle matter — adaptation costs about 1.4 points of score rate, everywhere.

7.3 P3: the oracle ablation — perfect classification worth exactly zero

Corollary 2 predicted not a small oracle gain but a structurally exact zero, and the measurement delivered it: in all nine cells, over 400 paired deals each, the oracle arm's per-deal delta against the classifier arm is exactly 0.0000 with SE exactly 0 (Table 5) — with a dominant row both arms delegate identically at every decision, so the paired games are identical move for move. **P3 is confirmed** — and what was confirmed is worth pausing on: a *perfect* classifier, an oracle no feature engineering could beat, buys nothing over the shipped one. Against this roster, the ceiling on the value of the entire observe-and-classify half of the architecture is the number in this table.

7.4 P4: classifier accuracy — mixed

The classifier was measured for its own sake across the 36 cross-play pairings (Table 6). Overall top-1 rises from 16.4% at 40 observed events to 20.2% at 80 and 22.4% on the full log, against a $1/9 = 11.1\%$ chance floor — real skill, roughly twice chance, and modest. Two structural notes: the 250-event checkpoint scored *zero* seats, because no public log outlived 250 events — the artifact records it as a *dead* checkpoint (`deadCheckpoints`), never as a measured zero; and the 150-event figure (37.5%) covers only the 672 surviving seats of the 6.2% longest games — a population selected toward the long-game styles, not comparable to the full-population rows.

Table 5: The oracle ablation: true styles handed to the adaptive seats vs. the classifier arm on identical seeds, 400 pairs per cell [14]. The two arms played byte-identical games.

Opponent	Classifier arm	Oracle arm	Δ (per-deal)
Balanced	0.5238	0.5238	0.0000 ± 0.0000
Blitz	0.5138	0.5138	0.0000 ± 0.0000
Punter	0.4975	0.4975	0.0000 ± 0.0000
Banker	0.5450	0.5450	0.0000 ± 0.0000
Turtle	0.6588	0.6588	0.0000 ± 0.0000
Hoarder	0.5775	0.5775	0.0000 ± 0.0000
Scout	0.6663	0.6663	0.0000 ± 0.0000
Ghost	0.5625	0.5625	0.0000 ± 0.0000
Archivist	0.5975	0.5975	0.0000 ± 0.0000

Table 6: Classifier top-1 by observed-event checkpoint, over 36 pairings $\times$ 50 games $\times$ 6 seats [14]; a checkpoint scores only games whose log outlived it. Chance is $1/9 \approx 0.111$.

Checkpoint	Seats scored	Top-1	Note
40 events	10,800	0.164	all games
80 events	10,464	0.202	nearly all games
150 events	672	0.375	the 6.2% longest games; selected population
250 events	—	—	dead: no public log reached 250 events
full log	10,800	0.224	the deployment-relevant figure

Table 7: End-of-game top-1 per true style, with each style's most frequent *prediction* and its share — the confusion attractors [14]. Chance is 0.111.

True style	Top-1	Most predicted (share)	Reading
Ghost	0.504	Ghost (0.504)	the one genuinely recognisable style
Scout	0.413	Scout (0.413)	recognisable
Hoarder	0.309	Hoarder (0.309)	recognisable via declare signatures
Turtle	0.272	Turtle (0.272)	recognisable via own-hand-only declares
Punter	0.129	Ghost (0.275)	quadrangle: absorbed by Ghost
Banker	0.126	Ghost (0.323)	quadrangle: absorbed by Ghost
Archivist	0.118	Scout (0.331)	absorbed by Scout
Blitz	0.093	Ghost (0.261)	quadrangle: absorbed by Ghost
Balanced	0.056	Ghost (0.270)	the control; below chance

Per style (Table 7), the pre-registered three-condition verdict came out **mixed**. The quadrangle condition holds: Balanced, Blitz, Punter and Banker classify at 0.056–0.129 (mean 0.101, condition ≤ 0.35) — exactly the near-indistinguishability the separability bound predicted. The distinctive-style condition fails: Turtle 0.272, Ghost 0.504, Hoarder 0.309 average 0.362 against the ≥ 0.50 bar. And the error-geometry condition fails informatively: quadrangle errors do not land inside the quadrangle (30–40% do, against the $\geq 50\%$ bar) — they land on *Ghost*, and Archivist's land on *Scout*. The confusion structure is not four styles mistaken for each other but a cloud of near-identical profiles absorbed by whichever fingerprint sits nearest in the pooled-z geometry; Balanced itself classifies *below* chance (0.056), its central fingerprint leaving almost any noisy draw closer to some other style's mean.

Table 8: The warmup tax, decomposed: warmup share w of the adaptive team's decisions, Balanced deficit g = (Punter row − Balanced row) in the opponent's column, prediction $-wg$, and the measured gauntlet delta [14].

Opponent	w	g	Predicted $-wg$	Measured Δ	$\lvert\text{diff}\rvert$/SE
Balanced	0.580	0.0190	−0.0110	-0.0076 ± 0.0045	0.76
Blitz	0.582	0.0236	−0.0137	-0.0112 ± 0.0058	0.44
Punter	0.578	0.0190	−0.0110	-0.0108 ± 0.0031	0.05
Banker	0.570	0.0199	−0.0113	-0.0076 ± 0.0067	0.56
Turtle	0.522	0.0185	−0.0096	-0.0097 ± 0.0065	0.00
Hoarder	0.551	0.0171	−0.0094	-0.0086 ± 0.0057	0.14
Scout	0.534	0.0226	−0.0121	-0.0092 ± 0.0065	0.44
Ghost	0.567	0.0319	−0.0180	-0.0177 ± 0.0069	0.05
Archivist	0.555	0.0265	−0.0147	-0.0148 ± 0.0066	0.01

7.5 The delegation record: the mechanism, observed

The gauntlet instrumented every adaptive decision with the style it delegated to. The record is total and unambiguous [14]: in all nine cells, 100% of warmup decisions delegated to Balanced (the anchor) and 100% of warm decisions delegated to Punter — not one of the 12.1 million warm decisions across the gauntlet chose anything else. This is Corollary 1 observed in play, at every posterior the classifier actually produced. The warmup phase accounted for 52.2% (vs. Turtle) to 58.2% (vs. Blitz) of the adaptive team's decisions. Warm = pure Punter plus warmup = a majority of decisions on a dominated style is the entire causal story of Tables 3 and 4, made quantitative next.

8 Discussion

8.1 Anatomy of the warmup tax

Why does a 40-event warmup swallow half the game? Because the public log grows far more slowly than the decision count: declining to declare emits no public event (Section 4.1), and declare-window declines dominate play. Gauntlet games average 683–767 engine moves, of which the adaptive team's three seats decide 342–384, yet across the 1,800 accuracy games — whose lengths are of the same scale — not one public log reached 250 events and only 6.2% reached 150 (Table 6). The warm threshold of 60 observed events therefore lands past the midpoint of the team's decisions: 52–58% of them are taken by the anchor.

A deliberately crude model prices this: suppose a game's score rate were linear in the mixture of styles played — warmup share w at the Balanced row's payoff, $(1-w)$ at the Punter row's — so the predicted shortfall against pure Punter is $-wg$, where g is the Balanced deficit in that opponent's column of Table 2. Table 8 runs this per cell: the zeroth-order prediction lands within 0.8 standard errors of the measured delta in all nine cells, and the ordering is reproduced — the worst measured cell (Ghost, −0.0177) is exactly the column where the Balanced-vs-Punter gap is widest (0.0319). Nothing subtle is happening: the tax is the anchor's known per-decision deficit times the fraction of decisions the anchor takes; the mixed screen's −0.0136 sits in the same range, as a deal-weighted average over 24 columns must.

8.2 The obvious remedies, and why each one is the finding

Two one-line configuration changes eliminate the tax, and both are instructive to refuse.

Set the anchor to Punter. Then warmup plays Punter, the warm argmax is Punter (Theorem 1, with the anchor bias now *reinforcing* the dominant row), and the "adaptive" policy is the static Punter with a classifier attached as ornamentation. **Set warmup to zero.** Then the earliest posteriors are the damping prior — near-uniform — and the argmax under uniform is still the dominant row: Punter from move one. Intermediate tunings behave the same way: shortening warmup, shrinking the phase, or raising the switch margin each moves the policy monotonically toward a static endpoint (always-anchor or always-Punter), because under dominance there is no belief at which the warm choice is anything else. Every knob that repairs the tax does so by deleting the adaptation.

This is not a criticism of the knobs; it is the result. Over a dominance-solvable roster, the best policy in the adaptive family is its degenerate member, and any non-degenerate member pays for its flexibility with a measurable loss.

8.3 What the classifier can and cannot see

The classifier's limits connect quantitatively to the inert-axis result [7]. The decision-divergence instrument of v0.5 measured how often each style's chosen action differs from the control's on identical views: Scout 2.89%, Ghost 2.15%, Archivist 1.66%, Banker 1.19%, Turtle 0.83%, Punter 0.47%, Hoarder 0.47%, Blitz 0.39% [11]. A public-log classifier sees only the accumulated consequences of divergent decisions, so styles that diverge on under 3% of decisions — all of them — leave thin trails, and the quadrangle styles, at under 1.2%, leave almost none: their 0.056–0.129 top-1 is the separability bound made flesh. But divergence *rate* is not the whole story: Hoarder and Turtle diverge rarely yet classify at 0.31 and 0.27, their rare divergences concentrated in declare events with loud public signatures (`foreignDeclareShare`, `ownHandOnlyShare`, `declareBackload`), while Archivist diverges three times as often yet classifies at 0.118, its trail absorbed by Scout's neighbouring fingerprint. What a behavioural classifier reads is not how often a style is different but how *loudly*, per event the log deigns to record. None of this matters to the payoff — Section 7.3 priced the whole apparatus at exactly zero against this roster — but it is the part of v1.0 that would survive a roster where classification did matter; the measured accuracies calibrate what it could contribute.

8.4 What would make adaptation valuable

The degeneracy is a property of the meta-game, not of the machinery, which was deliberately built so that a different meta-game flows through unchanged (its tests re-derive the argmax from the table rather than pinning Punter [15]). Three departures would give the classifier's output a decision to influence:

- **An intransitive roster.** Theorem 1 needs a dominant row. A roster engineered with genuine counters — which v0.5 measured this one not to be (cyclic energy 0.0112) [6] — makes the argmax belief-dependent, and then classifier accuracy and warmup cost become terms in a real trade-off.

- **Off-roster opponents.** Humans at the project's play table are the actual target of the S35/A14 program [9]: they are not columns of any committed table, their tendencies drift, and exploiting them requires exactly the observe-classify-respond loop built here — plus fingerprints calibrated from logged human play, a data-collection decision the synthesis flags as prior to everything else.

- **Rule-set shifts.** The `us54` gift rule is what prices Punter's risk appetite correctly [6]; under the 48-card baseline the declaration economics flip, the counter table would need re-measurement, and nothing guarantees the new table a dominant row.

Until one of those holds, the honest statement stands: over this roster, under these rules, the best adaptive policy is the degenerate one, and every measured alternative is worth less than nothing.

9 Threats to validity

1. **Cross-run pairing (P1).** The gauntlet's delta SEs are conservative upper bounds (Section 6.3) — which can only have hidden significance, not manufactured it; the truly-paired, seed-clustered P2 screen agrees at $z = -3.2$.

2. **Dominance is of the committed table.** The theorem consumes point estimates; 70 of 72 dominance comparisons clear the corrected bound, but the weakest margin (0.0112, Blitz row, Hoarder column) is only 1.86 conservative SEs from zero (Section 5). A true matrix inverting it would make the warm argmax belief-dependent in a narrow region, changing the result's form but not its economics (the value at stake is $\lesssim$0.011).

3. **Fingerprints are calibrated in mirror games.** Deployed reads happen in cross-play, where opponents shape game length and event mix. The checkpoint buckets and length-invariant features mitigate this, and the accuracy experiment itself measures on cross-play pairings, but the mismatch is real and untuned.

4. **Warmup and phase constants are design choices, not tuned values.** Section 8.2 is why tuning was not pursued: under dominance every tuning direction terminates in a static policy, so a tuned v1.0 would measure only the speed of its own disappearance.

5. **Statelessness is a design constraint with alternatives.** A stateful agent could implement hysteresis with memory and act mid-phase; Theorem 1 does not depend on the phase construction, but the measured warmup boundary (60 events) is an artifact of the stateless gate.

6. **Closed population.** All conclusions are self-play within one roster, one engine, one rule set (pinned by the artifact's rules hash); the transfer warning of [5] applies in full, and Section 8.4 lists the exits. The classifier-accuracy numbers are additionally conditional on full-strength opponents. (The adaptive mirror's exact 0.5000 ± 0.0000 certifies harness symmetry only; correctness rests on the engine's audits [6, 15].)

10 Conclusion

FishAI v1.0 built the adaptive agent the style debate keeps invoking — observe the table, identify the styles, counter them — and measured what it is worth against the measured meta-game. The answer has three layers. Adaptation over this roster *converges*: the counter table's Punter row dominates every column, expected payoff is linear in belief, and the warm agent delegated every one of twelve million measured warm decisions to Punter — the theorem observed in play. Adaptation over this roster *costs*: the warmup anchor plays a dominated style for 52–58% of decisions because the public log, which records no declines, matures far more slowly than the game, and the resulting tax — -0.0108 mean across the gauntlet, -0.0136 ± 0.0043 truly paired in mixed company — is reproduced within one standard error, in every cell, by the crudest possible mixing model. And classification, the part of the architecture with genuine measured skill (22.4% top-1 against 11.1% chance, half a coin-flip on the Ghost, structurally blind inside a quadrangle of styles that diverge on under 1.2% of decisions), is worth exactly zero: the oracle deltas are 0.0000 to the last digit, because under dominance no belief changes the play.

Two of four pre-registered predictions were refuted, both in the informative direction: the project predicted convergence and got it, predicted the convergence would be free and was billed for it. The billing is the contribution. "Just adapt" is, over this roster and under these rules, strictly worse than picking the best static style and playing it — worth less than nothing — and every configuration that repairs the loss does so by quietly becoming that static style. That is a measured fact about this meta-game, not a verdict on adaptation as an idea: an intransitive roster, an off-roster human, or a rule shift would each hand the machinery a real decision, and the machinery is ready for the first opponent that gives it one.

Reproducibility. The engine is deterministic; the artifact [14] carries the seed prefixes, configuration, digests, commits, rules hash, the verbatim predictions and both calibrations' provenance; every number here is in it, the counter table [13], or the measurement documents [10, 11, 12].

References

[1] D. Balduzzi, K. Tuyls, J. Pérolat, and T. Graepel. Re-evaluating evaluation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[2] S. Omidshafiei, C. Papadimitriou, G. Piliouras, K. Tuyls, M. Rowland, J.-B. Lespiau, W. M. Czarnecki, M. Lanctot, J. Pérolat, and R. Munos. α-Rank: Multi-agent evaluation by evolution. *Scientific Reports*, 9:9937, 2019.

[3] S. V. Albrecht and P. Stone. Autonomous agents modelling other agents: A comprehensive survey and open problems. *Artificial Intelligence*, 258:66–95, 2018.

[4] D. Rebstock, C. Solinas, M. Buro, and N. R. Sturtevant. Policy based inference in trick-taking card games. In *IEEE Conference on Games (CoG)*, 2019.

[5] H. Hu, A. Lerer, A. Peysakhovich, and J. Foerster. "Other-play" for zero-shot coordination. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2020.

[6] A. Hsieh. FishAI v0.5: Measuring play style without skill in a 54-card Literature variant. FishAI project; companion paper, published alongside this one, 2026.

[7] A. Hsieh. The inert axis: when a style parameter is wired, swept, and never reached. FishAI project; companion paper, published alongside this one, 2026.

[8] A. Hsieh. The contained book: an absorbing resource measured to be worth nothing. FishAI project; companion paper, published alongside this one, 2026.

[9] FishAI project. PLAYSTYLES.md — Literature / Canadian Fish: play styles, rule dialects, and engine specification. Internal research synthesis, 2026. Part V catalogues the opponent-modelling families (A14; the S35 detection panel) this paper implements a slice of; its unverified secondary citations are not relied on here.

[10] FishAI project. RULES_US54.md — the "US student" 54-card variant. Repository rules specification with derived consequences and test vectors, 2026.

[11] FishAI project. STYLES.md — the play-style roster under `us54`. Repository document; §6.1 the inert axis and decision-divergence figures, 2026.

[12] FishAI project. BOT_LAB.md — play-style spectrum: research, experimental design, and metrics. Repository document; §4.3 health gates, §5 experimental design, 2026.

[13] FishAI project. Tournament artifact `src/lab/data/style-results.v2.json` and the derived counter table `lib/engine/bots/data/counter-table.ts`. 36 cells × 4,300 duplicate pairs; records digest `0c3cb524a3534d4a`; engine commit `1667a1d` (recorded `1667a1d-dirty`); generated 2026-08-22.

[14] FishAI project. Adaptive artifact `src/lab/data/adaptive-results.json`. 125,600 games; records digest `ce3316641a8f38fe`; engine commit `45ec9f3` (recorded `45ec9f3-dirty`); rules hash `b4c7fef7...`; generated 2026-08-24.

[15] FishAI project. FishAI repository: engine, bots, simulation laboratory, and results site. Adaptive layer: `lib/engine/bots/observe.ts`, `classify.ts`, `adaptive.ts`; experiment suite: `lib/lab/adaptive.ts`, 2026.